A Testing Scheme for Mixed-Control Based Reversible Circuits

Bappaditya Mondal*, Chandan Bandyopadhyay† and Hafizur Rahaman†
*Department of Computer Science & Engineering, Neotia Institute of Technology Management and Science, India
†Department of Information Technology, Indian Institute of Engineering Science and Technology, Shibpur, India
E-mail: bappa.arya@gmail.com, chandanb.iiest@gmail.com, hafizur@it.iiests.ac.in

***Abstract*— In recent years, the design and synthesis of the reversible quantum circuits have received immense importance in computing world. As the advancement is going on, simultaneously newer challenges are evolving like fault detection and testing in the reversible circuits. To address these issues, proper testing techniques will have to be developed for efficient detection of the faults. In this work, we propose a fault detection scheme for efficient detection of the faults in any type of reversible circuits which may consist of completely positive or completely negative or mixed controlled Toffoli gates. This is suitable for very large circuits with minimum number test vectors. We have tested our scheme on several reversible benchmark circuits. This scheme detects fault as well as localizes the faulty zone. Experimental results are presented and a comparative analysis with recent works has been made.**

Keywords: - Reversible logic; mixed polarity; test vector; missing gate fault.

I. INTRODUCTION

Quantum computing [1-2] is one of the most promising computing technologies in recent days. In the recent years, several remarkable research works have been reported in this field and the practical realization of such circuit has been initiated [3-6]. In a recent breakthrough, 2-qubit Quantum gate has been developed in Silicon [7].

Various synthesis techniques have been reported in [10-13], but there is no direct technique for the synthesis of the quantum circuits [8-9]. Actually, the boolean functionality of quantum circuits is synthesized into the reversible circuit. Then the reversible circuits are mapped to the quantum circuits. Another important issue is testing of reversible quantum circuits. Several works have come out [14-16], where the investigations of the detection and localization of faults in reversible circuits have been discussed.

Use of deterministic and probabilistic tests to detect faults in reversible quantum circuit has been discussed in [16]. In [14], an integer linear program *(ILP)* based technique is used to minimize the test set for detection of multiple stuck-at faults in the reversible circuits. Furthermore, this work has been extended in [23], where the optimal test sets (*OTS*) are generated for all type of missing gate faults in the reversible circuit.

A testing approach has been developed in [20] which is named as 'Ping Pong test' that achieves 100% fault coverage. The fault localization in the reversible circuits is easier than that in the classical circuits and this fact is established in [17], where an adaptive tree based technique has been developed to establish the fact.

An automatic test pattern generation for the fault detection in reversible circuits has been introduced in [18]. In [19], a design technique has been incorporated to detect all type of missing gate faults, where one extra control line along with the duplication of each *k-CNOT* gate has been added in the testable design.

Another testing approach based on boolean difference has been reported in [21]. The design-for-testability (*DFT*) based [22] testing techniques have been developed recently. All the testing techniques discussed so far detect faults in the reversible circuits implemented with only positive control Toffoli gates. But if we consider the reversible circuits implemented with only negative control Toffoli (*n*-Toffoli) gates or mixed control based Toffoli (*m*-Toffoli) gates, then the existing testing techniques may not be effective.

In this work, we have developed a testing scheme to detect faults like *SMGF* and *PMGF* in the reversible circuits constructed with any type of Toffoli gates. The developed testing scheme not only efficiently finds the mentioned faults in the reversible circuits by producing lesser number of the test vectors but also ensures the detection of the faults in the large circuits as well. To this end, the experimental results with the comparative analysis are also given to justify the proposed scheme.

The rest of the work is organized as follows. The proposed technique is explained in Section II. Comparative study with existing works is presented in Section III and finally the conclusion remarks appear in Section IV.

Preliminaries:

Definition 1: *A circuit can be said a reversible one if only if the mapping from input to output is bijective and the number of inputs (m) is equal to number of outputs (n) i.e m = n.*

In any reversible circuit, every component performs reversible operation. Basically, the reversible circuits are designed using reversible gates, which operate over set of control and target lines. Toffoli [24], Feynmen [25], Fredkin [26] are the examples of such reversible gates but mainly it has been found that the most of the reversible circuits consist of Toffoli gates. Such Toffoli gates are classified into three types based on the nature of control nodes and they are *positive controlled* Toffoli *(p*-Toffoli*)*, *negative controlled*

978-1-5090-2541-1/16/$31.00 ©2016 IEEE

Toffoli *(n*-Toffoli*)* and *mixed controlled* Toffoli *(m*-Toffoli*)*.

Definition 2*: If all the controls in a Toffoli gate are positive in the polarity, then the gate is termed as positive controlled* Toffoli*. If all the controls are negative then negative controlled* Toffoli *or if the controls are combinations of positive and negative then it is called mixed controlled* Toffoli gate*.*

Example 1: The design of a 2-controlled Toffoli gate and its *NCV* quantum gate library based realization is given in Fig. 1(a) and Fig. 1(b), respectively. In Fig. 2, all the various types of Toffoli gates are shown. An example of the reversible circuit has been shown in Fig. 3, where the circuit is designed with such Toffoli gates having various control types.

A circuit can be said a faulty if it does not produce the exact output value as it is desired. This fault may be generated due to the missing of the gates or due to the missing of the control nodes in the circuit. Depending upon this nature of faults, the faults in the reversible circuits can be categorized into three classes: Single Missing Gate Fault (*SMGF*), Partial Missing Gate Fault (*PMGF*) and Multiple Missing Gate Faults (*MMGF*). Such faults can be detected by applying test vectors. Such test vectors contain a certain boolean patterns, when they are applied over a certain circuit it checks the circuit correctness by judging the applied input pattern and the obtained output patterns.

Definition 3**:** *If a single gate completely disappears from the circuit then this type of fault is known as Single Missing-Gate Fault.*

This fault can be detected by applying '1' at the respective control lines of the missing gates and setting the target line as well as all through lines to either '0' or '1'.

Definition 4**:** *When two or more consecutive gates disappear from the circuit, then such type of fault is termed as multiple missing-gate faults (MMGF).*

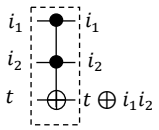

Fig. 1 (a): 2-ControlToffoli gate

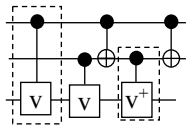

Fig. 1 (b): Equivalent NCV quantum gate library based realization of Fig. 1(a)

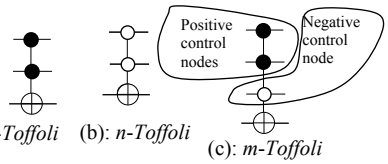

Fig. 2: Classification of Toffoli gates based on control types

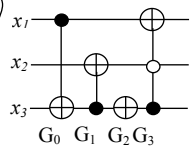

Fig. 3: Example of reversible circuit

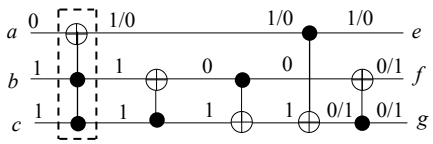

Fig. 4 (a): *SMGF* in *ham3ndesign#1*

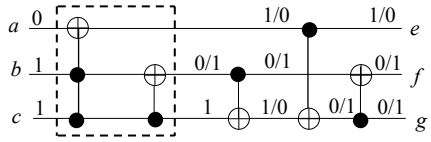

Fig. 4 (b): *MMGF* in *ham3ndesign#1*

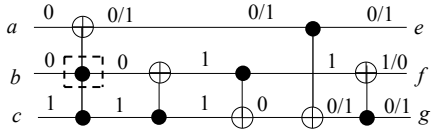

Fig. 4 (c): *PMGF* in *ham3\design#1*

For such type of faults the selection of test vectors depends on the type of faults as well as positions of faults. Similarly, for the circuit given in Fig. 4(b), where the first two gates are missing and to detect the said fault the applied test vector will be <011>.

Definition 5**:** *A partial missing-gate fault occurs in reversible circuit if the control nodes of any gate go missing.*

To detect such faults, '0' value is set to the respective control line in which the control node of the gate is missing and '1' value is applied to other control lines and the target line as well as all other through lines are connected to either '0' or '1'.

Example 2: Let's consider the benchmark circuit *ham3ndesign#1*. Missing of gates or controls is marked by the dotted box in the diagram.

In the Fig. 4(a), single missing gate fault occurs due to the missing of the first gate from the circuit. If <011> test vector is applied at the input of the circuit, the output would show <011> pattern instead of normal circuit output which is <100> due to this *SMGF*. Hence, the *SMGF* marked by the dotted box fault is detected.

In the circuit shown in Fig. 4(b), a *MMGF* occurs due to the missing of two successive gates marked by the dotted box. The test vector <011> detects the *MMGF* fault.

In Fig. 4(c), the control at *b* line goes misses from the first gate causing *PMGF* in the circuit. The applied test vector <001> detects this missing control fault in this case.

II. PROPOSED TECHNIQUE

Here, we are presenting techniques for detecting two faults in reversible circuits:- *SMGF* and *PMGF*. This technique detects the faults in the reversible circuit implemented not only with positive control based Toffoli gates but also with mixed-control based Toffoli gates. This technique is also applicable to highly scalable circuits and needs lesser number of test vectors.

The complete process of fault detection has three phases - *circuit augmentation phase, fault localization phase* and *fault detection phase*. Here, we are explaining all these phases sequentially with a model circuit.

Let, consider the design of Fig. 5(a) as shown in the Fig. 5(a).

This circuit is implemented with 3 input lines and 4 multiple control Toffoli (*MCT*) gates. Now, this circuit is to be tested for *SMGFs* or *PMGFs*.

A. Circuit Augmentation Phase:

In this augmentation phase, we add a set of Toffoli gates and one additional line to the circuit to make the circuit testable. This process is completed in three steps:

Step1: Add an additional line (L_a) and set to 1 at L_a.
Step2: Add one *CNOT* gate before the negative control node and after the negative control node as shown in Fig. 5(c). This step converts the mixed polarity circuits into positive polarity circuit.
Step3: For a circuit having *N* number of *MCT* gates, add *N* additional positive polarity gates in the circuit, whose one of the control is connected with line L_a.

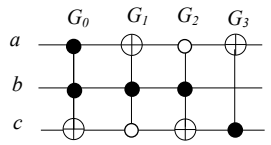

Fig. 5 (a): Input model circuit (C_{in})

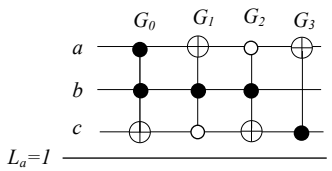

Fig. 5(b): Modified circuit after adding one line with Fig. 5(a) as *step1*

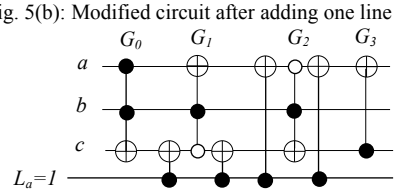

Fig.5 (c): Updated circuit after executing *step2* over Fig. 5(b)

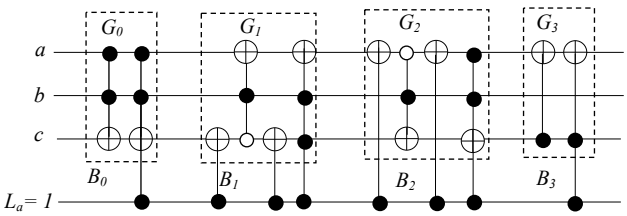

Fig.5 (d): Circuit obtained after completion of *Circuit Augmentation Phase*

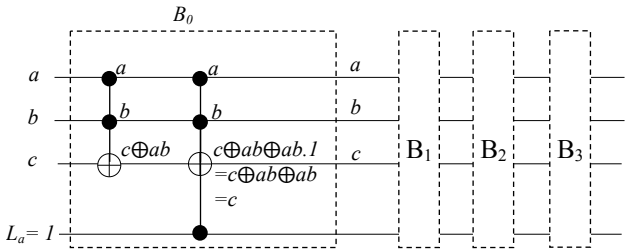

Fig.5 (e): Block B_0 of Fig. 5 (d) produces same input and output vector

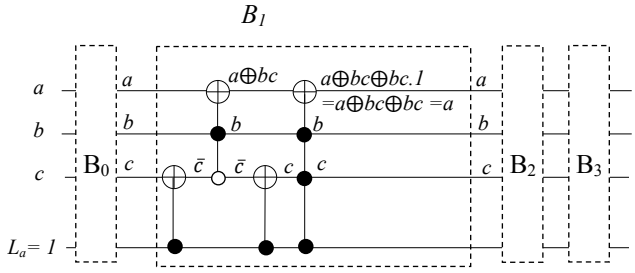

Fig.5 (f): Block B_1 of Fig. 5 (d) produces same input and output vector

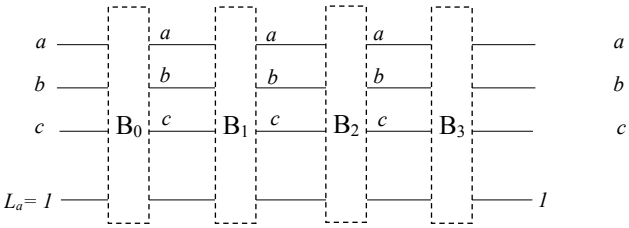

Fig.5 (g): Each block of Fig. 5 (d) produces same input and output vector

Now, the circuit of Fig. 5(a) has been transformed into a testable circuit as shown in Fig. 5(d). But if the input value of additional line L_a is set to 0, then this testable circuit behaves as original circuit of Fig. 5(a) and this modification maintains its original functionality.
But when L_a is set to 1, then the patterns applied at the primary input of the circuit remain unaltered at the primary output of the circuit. In the augmented circuit, the circuit is decomposed into several blocks and the output of each block remains unchanged. The circuit is augmented in such a way that each augmented block passes the exact input pattern to the output of that block. In Fig. 5(e) and Fig. 5(f), we have shown that the first two blocks B_0 and B_1 of Fig. 5(d) keep the vectors unchanged and similarly we can say that the rest of the blocks do the same as block B_0 and B_1 do and keep the input vector unchanged.

B. Fault detection phase:

In the last section, we have shown how an input circuit is transformed into a testable one (augmented circuit) by adding redundant gates and one ancille line to the input circuit. In this phase, we only check whether a circuit is faulty or not and in next phase (*Fault localization phase),* we will locate the exact positions from where the faults are originated in the circuit.

In practical case, the test patterns are passed through the circuit and it is checked whether the circuit under test (*CUT*) is faulty or fault-free. If the output response is same as the predicated output, the *CUT* is declared as fault-free circuit; otherwise, it is a faulty circuit.

But the test generation is an important phase in testing of a circuit. Now, here we are stating the test vector generation method in details.

Generation of Test vector: In the proposed testing technique, one test vector is required to test all *SMGFs* and *n* test vectors is sufficient to detect *PMGFs* in any arbitrary circuit having *n* circuit lines.
In an *n* variable circuit consisting of *n* circuit lines, all the

possible *SMGFs* can be detected if the applied input test vector is <11…1> i.e. *n* number of 1s are applied at the input of the circuit. It does not depend on numbers of gates present in this circuit. When the applied test vector <11…1> passes through the blocks, the pattern does not changes which ensures detection of SMGF in the circuit. So, only the test vector <11…1> is sufficient to detect any *SMGF* in the circuit. But this test vector cannot detect *PMGFs*. Circuit consisting of *n* circuit lines and *m* reversible gates needs *n* different test vectors to detect *PMGFs*. The first test vector is <011…11> and if the bits in the test vector are shifted right then a new test vector will be generated and for a *n*-bit test vector, total *n* number of test patterns will be generated, where there will be (*n*-1) number 1s and only one 0 in each pattern. For the circuit of Fig. 5(d), the test vector <111> is required to detect *SMGFs* and the test vectors {<011>, <101>, <110>} are sufficient to detect the first order *PMGFs*.

Fault Detection Mechanism: To check the circuit whether it is faulty or fault-free, first, the circuit is augmented into a testable circuit. Then, all the test vectors will be applied to detect the faults. After applying the specified test vectors, if it is found that the input test pattern is not same as the output test pattern, then the circuit will be declared as faulty circuit.

For example, consider Fig. 5(d) whose third gate (G_2) is missing resulting an *SMGF* in the circuit. After removal of that gate from that circuit, a new structure is obtained in Fig. 6(a). Now, if we apply the test pattern <111>, we will obtain the output vector <110> from the circuit. As both the input and output patterns differs, so it can be concluded that the circuit is faulty due to the missing of the gate G_2.

Here, we have shown the detection of missing of *m*-Toffoli gate from the circuit. Similarly, the missing of any type of Toffoli gates (*p*-Toffoli, *n*-Toffoli) is also possible by the proposed testing technique.

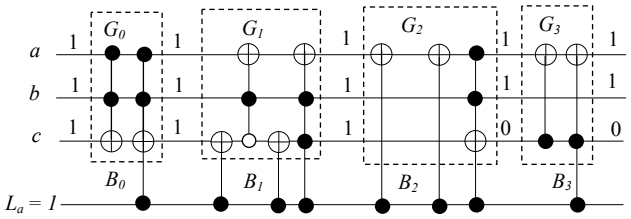

Fig. 6 (a): Faulty circuit due to *SMGF* in G_2 of Fig. 5(d)

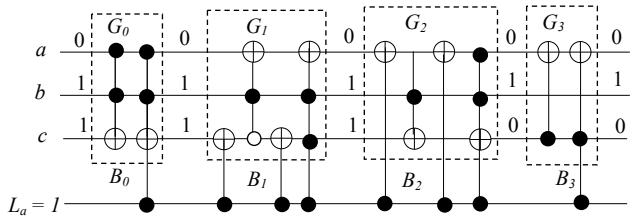

Fig. 6 (b): Faulty circuit due to missing of negative control (*PMGF*) in G_2 of Fig. 5 (d)

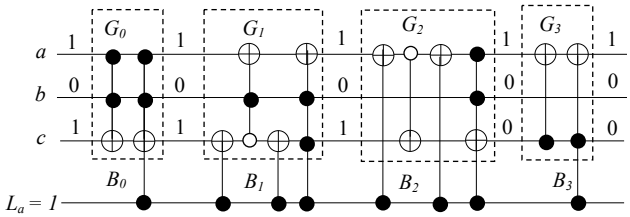

Fig. 6 (c): Faulty circuit due to missing of positive control (*PMGF*) in G_2 of Fig. 5 (d)

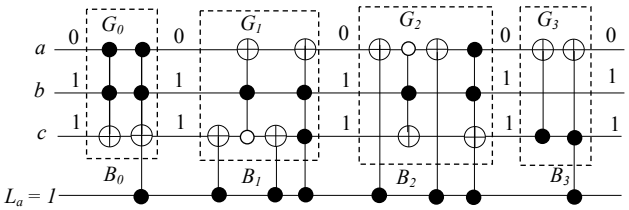

Fig. 6 (d): Fault free circuit shows input equal to output

Again if we consider in spite of one gate missing from the circuit, a control misses from a certain gate and let us consider that the negative control in gate G_2 of block B_2 is missing as shown in Fig. 6(b). As like previous case, now we apply test pattern <011> and we can see that its output <010> also differs from the input vector which confirms the presence of the fault.

Similarly, if a positive control misses in that gate, then the test vector <101> detects the fault as shown in Fig. 6(c). Hence, the proposed testing technique can efficiently detect missing of negative control as well as the positive control.

But if the circuit does not contain any faults then application of any of the test vectors never changes the output patterns as the circuit is fault-free and this fact is shown in Fig. 6(d).

C. *Fault localization phase:*

In previous section, we have discussed how to check whether a circuit is faulty or not. But one important issue that still remains is localization of fault. Without localization of the fault, the circuit cannot be repaired and also not possible to restore to its original.

To find the faulty area, we have followed a simple approach where we have scanned the entire circuit block wise from left (input section) to right (output end) and have compared the input and output vector block wise. If we find any of the blocks output do not match the said vector patterns, then it can conclude that the block contains the faulty gate.

For example, the circuit shown in Fig. 6 (a) contains some error due to missing of a gate and the presence of that fault is already confirmed in previous phase. Now if we perform block wise checking on that circuit, then it can be found that the input-output vectors patterns (<111> and<110>) only differs for block B_2. So, it can be decided that the fault is originated from block B_2.

Similarly, by applying the same approach in Fig. 6 (b) and Fig. 6 (c) the fault is tracked to block B_2 and block B_3.

III. EXPERIMENTAL RESULTS

The comparison of the results is given in Table 1, where we have compared our scheme with some recent related works reported in [18], [20]. From the table, it can be seen that our technique takes lesser number of test vectors to detect the faults and this improvement is quite high. Another feature in our testing technique is that it can deal with any type of circuits whereas algorithms in [18] or [19] or [20] only can take a special class of circuits implemented with positive control *MCT* gates only.

IV. CONCLUSION

This work has presented a testing scheme to test any type of reversible circuit. First time, the testing of reversible circuit implemented with mixed control *MCT* gates has been proposed. This scheme needs lesser number of test vectors to test mixed control reversible circuit though some additional circuit overhead is required to make the circuit testable. Presently, we are working over it to reduce the overhead and trying to include higher order *PMGFs* in our testing scheme.

TABLE I. COMPARATIVE ANALYSIS BASED ON NUMBER OF TEST VECTOR FOR SMGF AND PMGF

Circuit	*n*	Technique in [18]		Technique in [20]	Proposed Technique	
		SMGF	*PMGF*	*SMGF*	*SMGF*	*PMGF*
hwb7tc	7	13	24	14	1	7
hwb6tc	6	8	15	8	1	6
ham7tc	7	4	4	4	1	7
mod5a-dders	6	3	6	3	1	6
6symd2	10	2	3	2	1	10
rd73d2	10	3	4	3	1	10
5mod5tc	6	1	5	1	1	6
hwb4tc	4	2	5	2	1	4
rd53d2	8	2	3	2	1	8
rd53d1	7	2	8	2	1	7
2of5d2	7	2	3	3	1	7
mod5d2	5	1	2	1	1	5
mod5d1	5	1	2	1	1	5
317tc	3	2	2	2	1	3
ham3tc	3	2	3	2	1	3

ACKNOWLEDGMENT

This work has been supported in part by a Senior Research Fellowship awarded to Chandan Bandyopadhyay by the Council of Industrial and Scientific Research (CSIR), Govt. of India. Authors would also like to acknowledge the CTS innovation lab and TEQIP Phase II for providing some infrastructural support.

REFERENCES

[1] P. W. Shor, "Algorithms for quantum computation: discrete logarithms and factoring,"*Foundations of Computer Science*, pp. 124–134, 1994.

[2] L. K. Grover, "A fast quantum mechanical algorithm for database search,"*Theory of computing*, pp. 212–219, 1996.

[3] H. Haffner et al, "Scalable multiparticle entanglement of trapped ions,"*Nature*, pp. 643–646, 2005.

[4] M. Laforest et al., "Using error correction to determine the noise model,"*Physical Review A*, pp. 133–137, 2007.

[5] B. Kane, "A silicon-based nuclear spin quantum computer,"*Nature*, pp. 133–137, 1998.

[6] J. Ghosh, J. et al. "High-fidelity CZ gate for resonator based superconducting quantum computers,"*Physical Review A*, 2013.

[7] M. Veldhorst et al., "A Two Qubit Logic Gate in Silicon," *Nature*, 2014.

[8] A. Peres,"Reversible logic and quantum computers,"*Physical Review A*, pp. 3266–3276, 1985.

[9] R. C. Merkle, "Reversible electronic logic using switches,"*Nanotechnology*,pp. 21–40, 1993.

[10] M. Saeedi, M. S. Zamani, and M. Sedighi,"On the behavior of substitution-based reversible circuit synthesis algorithms: Investigation andimprovement,"*International Symposium on VLSI*, pp.428–436, 2007.

[11] P. Gupta, A. Agrawal, and N. K. Jha,"An algorithm for synthesis of reversible logic circuits,"IEEE*Transaction on Computer Aided Design*, pp. 2317–2329, 2009.

[12] D. Maslov, G. Dueck, and M. Miller,"Fredkin/toffoli templates for reversiblelogic synthesis," *International Conference on Computer-AidedDesign*, pp. 256–261, 2003.

[13] V. V. Shende, A. K. Prasad, I. L. Markov, and J. P. Hayes,"Synthesis of reversible logic circuits,"IEEE*Transaction on Computer Aided Design*, pp. 723–729, 2003.

[14] K. N. Patel, J. P. Hayes, and I. L. Markov,"Fault testing for reversible circuits," IEEE *VLSI Test Symposium*, pp. 410–416, 2003.

[15] J. P. Hayes, I. Polian, and B. Becker, "Testing for missing-gate faults in reversible circuits," IEEE *Asian Test Symposium*, pp. 100–105, 2004.

[16] M. Perkowski, J. Biamonte, and M. Lukac,"Test generation and fault localization for quantum circuits,"*International Symposium onMulti-Valued Logic*, pp. 62–68, 2005.

[17] K. Ramasamy, R. Tagare, E. Perkins and M. Perkowski, "Fault localization in reversible circuits is easier than for classical circuits," IEEE *Asian Test Symposium*, 2004.

[18] I. Polian, J. P. Hayes, T. Fienn, and B. Becker,"A family of logical fault models for reversible circuits," IEEE *Asian Test Symposium*, pp. 422–427, 2005.

[19] H. Rahaman, D. K. Kole, D. K. Das, B. B. Bhattacharya, "On the detection of missing-gate faults in reversible circuits by a universal test set," IEEE *International Conference on VLSI design*, pp. 163-168, 2008.

[20] M. Zamani, M. B. Tahoori, and K. Chakrabarty, "Ping-pong test:Compact test vector generation for reversible circuits," IEEE *VLSI Test Symposium*, pp. 164–169, 2012.

[21] B.Mondal, D.K. Kole, H.Rahaman and D. K. Das, "Generator for Test Set Construction of SMGF in Reversible Circuit by Boolean difference method," IEEE*Asian Test Symposium*,pp.68-73, 2014.

[22] J.Mondal, D. K. Das, D.K.Kole, H.Rahaman and B.B. Bhattacharya, "On Designing Testable Reversible Circuits Using Gate Duplication," *International Symposium on VLSI Design and Test*,pp. 322-329, 2013.

[23] D. K. Kole, H. Rahaman, D. K.Das and B. B. Bhattacharya, "Derivation of Optimal Test Set for Detection of Multiple Missing-Gate Faults in Reversible Circuits,"IEEE *Asian Test Symposium*, pp.33-38, 2010.

[24] T. Toffoli, "Reversible computing,"*MIT Lab for Computer Science*, *Technical memo MIT/LCS/TM-151*, 1980.

[25] R. Feynman, "Quantum mechanical computers," *In Foundations of physics*, pp. 507–531, 1986.

[26] E.Fredkin and T.Toffoli, "Conservative logic,"*International Journal of Theoritical Physics*, pp. 219–253, 1982.